WEEK 07 TECHNICAL REPORT

YCB-V CNN Classification I — CNN baseline on Bop object crops

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Group	Group 1 — Basic ML Knowledge
Main Quest	Convolutional Neural Networks
Side Quest	CNN Classification I

1. Objective and executive summary

The goal of Week 07 was to establish a clean convolutional neural network (CNN) recognition baseline on the same five-class YCB-V BOP'19 object subset used in Weeks 03-06. To isolate the architecture change more clearly, the run reused the Week 06 X2 preprocessing recipe: an 8% padded bounding-box crop followed by aspect-ratio-preserving, gray-padded resizing to 64 x 64 pixels. The same deterministic scene-based train/validation/test split was reconstructed. A compact CNN was trained, its checkpoint selected using validation loss alone, and that frozen checkpoint evaluated on 450 held-out object instances.

Key measured outcomes

HELD-OUT ACCURACY	MACRO-F1	GAIN VS WEEK 06 X2
72.67%	**0.7132**	**+9.78 pp**

The CNN correctly classified 327 of 450 held-out instances, versus 283 of 450 for the Week 06 X2 MLP. The macro-F1 rose from 0.6468 to 0.7132 (+0.0664). These results support the CNN as a stronger overall baseline under matched input geometry, but the gains are not uniform across classes.

Central finding

The remaining weakness is master_chef_can: only 18 of 75 samples were correctly recognized (24% recall), while 57 were predicted as tomato_soup_can. Sugar_box and tuna_fish_can reached 100% recall. A 14.0-percentage-point validation-to-test accuracy drop remains a warning about generalization to different scenes.

INTERPRETATION NOTE The internal 450-instance test split was formed inside the BOP'19 test-source archive. The reported accuracy is an internship-specific scene-held-out classification result, not an official YCB-V or BOP challenge benchmark.

2. Dataset and experimental protocol

The executed notebook loaded the Hugging Face bop-benchmark/ycbv archive pair ycbv_base.zip and ycbv_test_bop19.zip, then built instance-level examples from each scene's scene_gt.json and scene_gt_info.json. Only the five previously selected objects were used: tomato_soup_can (ID 4), sugar_box (3), master_chef_can (1), tuna_fish_can (6), and bleach_cleanser (12).

2.1 Scene-aware split and instance counts

Partition	Scenes (IDs)	Tomato	Sugar	Master	Tuna	Bleach	Total
Train	48, 49, 50, 53, 54, 56, 57, 58	202	225	150	150	150	877
Validation	52, 55	75	75	75	75	75	375
Final test	51, 59	150	75	75	75	75	450

No scene IDs overlapped across splits (8 / 2 / 2 scenes). Filtering required a valid object bounding box, width and height of at least 24 pixels, and visible fraction of at least 0.35. The notebook also imposed per-class sample caps and fixed the random seed to 42.

2.2 Input processing

- Extract the RGB crop from the object bounding box with 8% padding on each axis, clipped to image bounds.
- Resize using aspect-preserving containment within 64 x 64 pixels, centering the result on a gray RGB (127, 127, 127) canvas.
- Convert to an RGB tensor and normalize each channel with mean 0.5 and standard deviation 0.5; no augmentation was added in this controlled Week 07 run.



Figure 1. Ten example training crops after 8%-padded extraction and 64 x 64 letterboxing, taken from the executed notebook.

INTERPRETATION NOTE Because the classifier operates on ground-truth object crops, the reported task is cropped-object classification. Bounding-box detection, localization, and segmentation are out of scope for Week 07.

3. CNN architecture and training

3.1 Implemented CompactCNN

Stage	Operation	Output intent
Block 1	Conv2D 3 -> 32 (3 x 3, pad 1), ReLU, MaxPool(2)	Local color/edge features; 64 -> 32 spatial size
Block 2	Conv2D 32 -> 64 (3 x 3, pad 1), ReLU, MaxPool(2)	Richer features; 32 -> 16 spatial size
Block 3	Conv2D 64 -> 96 (3 x 3, pad 1), ReLU, adaptive average pool (4 x 4)	Spatial features reduced to 96 x 4 x 4
Head	Flatten, Linear 1536 -> 128, ReLU, Dropout 0.30, Linear 128 -> 5	Five unnormalized class logits

The implemented network contains 272,165 trainable parameters. Convolution preserves neighborhood structure within the input image, unlike flattening all pixels at the input of the older MLP. The five logits were optimized using cross-entropy loss; softmax probabilities were applied during inference.

3.2 Fixed training configuration

Optimizer / loss	Learning rate	Weight decay	Batch	Maximum epochs	Checkpoint rule
Adam / cross-entropy	0.001	0.0001	64	15, patience 4	Minimum validation loss

The executed run used GPU acceleration (CUDA; PyTorch 2.11.0+cu130) and trained for eight epochs before early stopping. Total logged training time was 102.2 seconds. The selected checkpoint was epoch 4, with validation cross-entropy 0.343059. At that checkpoint the validation accuracy was 86.67% and macro-F1 was 0.8611.

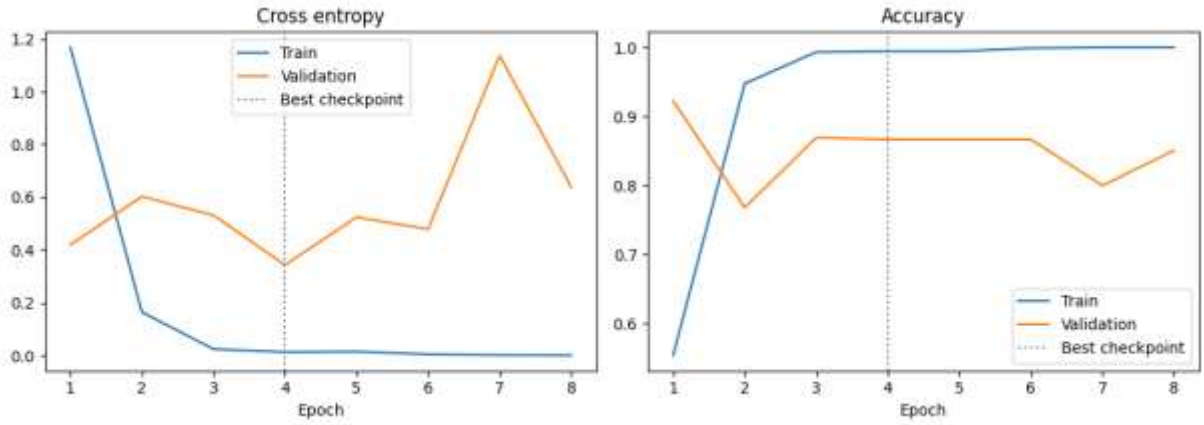


Figure 2. Train/validation loss and accuracy over eight epochs; the dotted line marks the epoch-4 checkpoint selected by minimum validation loss.

INTERPRETATION NOTE The first epoch had higher validation accuracy (92.3%) but worse validation loss (0.4201). The predeclared selection rule used loss, not peak accuracy; reporting the epoch-4 outcome avoids changing the rule after seeing results.

4. Held-out evaluation and class-specific performance

After checkpoint selection, the notebook ran one final evaluation on 450 object instances from scenes 51 and 59. Of those, 327 predictions were correct and 123 were incorrect.

4.1 Main metrics

Metric	Validation (375)	Final test (450)	Interpretation
Accuracy	86.67%	72.67%	-14.00 percentage points on held-out scenes
Macro-F1	0.8611	0.7132	-0.1479 absolute F1
Weighted F1	Not reported	~0.70	Reflects unequal support of tomato class

4.2 Per-class final test metrics

Class	Support	Precision	Recall	F1-score
tomato_soup_can	150	0.5659	0.7733	0.6535
sugar_box	75	0.8242	1.0000	0.9036
master_chef_can	75	1.0000	0.2400	0.3871
tuna_fish_can	75	0.8065	1.0000	0.8929
bleach_cleanser	75	1.0000	0.5733	0.7288

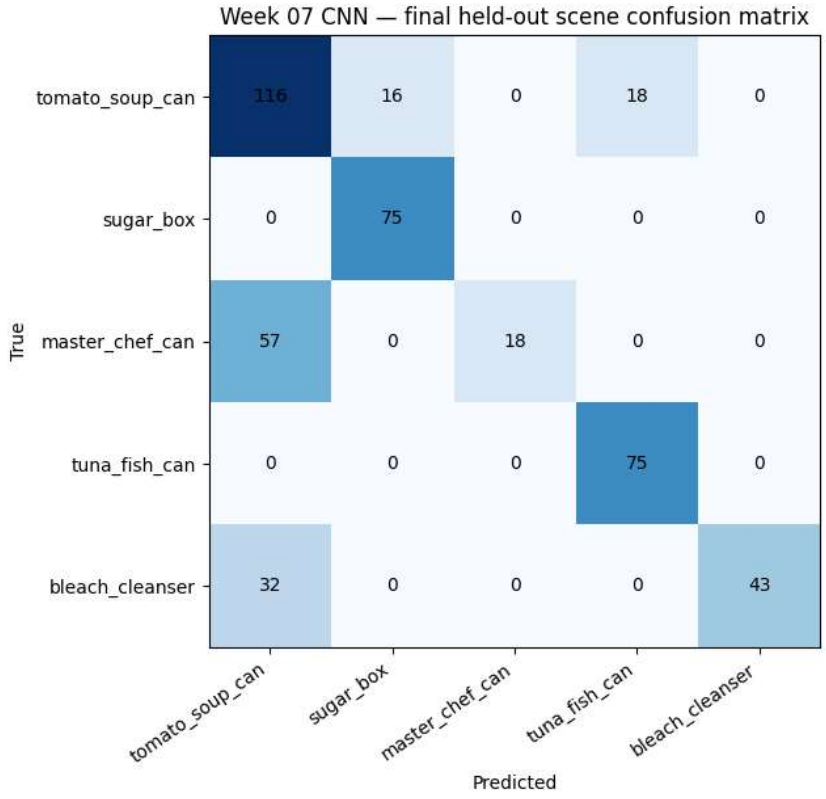


Figure 3. Final held-out test confusion matrix (true classes in rows; predictions in columns).

The overall accuracy conceals strong class asymmetry: perfect recall for sugar and tuna contrasts with only 24% recall for master_chef_can. Consequently, macro-F1 (0.7132) is an essential complement to accuracy, not a redundant secondary metric.

5. Error analysis

5.1 Dominant error transitions

Ground truth -> prediction	Count	Evidence / implication
master_chef_can -> tomato_soup_can	57	76% of master_chef_can objects were lost to the tomato class.
bleach_cleanser -> tomato_soup_can	32	Bleach retains 57.33% recall and is frequently absorbed by tomato.
tomato_soup_can -> tuna_fish_can	18	A subset of tomato instances is confused with tuna.
tomato_soup_can -> sugar_box	16	A smaller tomato-to-sugar confusion remains.

All 123 test errors are accounted for by these four directed transitions. The most striking error concentration is 57 master-to-tomato mistakes (46.3% of the 123 errors). This sharply lowers master recall even though the master predictions that were made are all correct (precision = 1.00).

5.2 Qualitative examples



Figure 4. The first eight incorrect crops in deterministic test-record order, directly from the executed notebook. Labels show true (T) and predicted (P) classes.

The shown failures include repeated tomato/tuna and bleach/tomato examples. Several near-identical crops appear because the gallery takes the first eight mistakes in record order, rather than sampling diverse failures by appearance or scene. It should be read as an inspectable example sequence, not as a representative random sample of all mistakes.

5.3 Diagnostic interpretation

Possible explanations include sensitivity to object appearance, pose, illumination, and background context across scene IDs, but the uploaded notebook does not isolate these factors experimentally. The train/validation curves also indicate a generalization risk: training accuracy approaches 100% while validation loss rises after the selected checkpoint. Neither observation alone proves a particular root cause.

6. Comparison with previous weeks

The historical table below is preserved for numerical context. The Week 07 CNN and Week 06 X2 MLP are the closest comparison because both use 64 x 64 aspect-preserving letterbox crops, the same five labels, and the same scene-held-out test split. A separate historical bar chart is intentionally omitted.

System	Test accuracy	Macro-F1	Preprocessing
Week 03 MLP	51.33%	Not recorded	Direct 64 x 64 resize
Week 04 E1 MLP	41.78%	0.4224	Direct 64 x 64 resize
Week 06 X2 MLP	62.89%	0.6468	64 x 64 letterbox
Week 07 CompactCNN	72.67%	0.7132	64 x 64 letterbox

Compared with Week 06 X2, Week 07 increased test accuracy by 9.78 percentage points (+44 correct predictions: 327 vs. 283) and macro-F1 by approximately 0.0664. The controlled preprocessing choice reduces one source of confounding, but the comparison is across separately executed training runs rather than a multi-seed ablation; it should not be interpreted as an isolated causal estimate of the CNN advantage.

7. Conclusion and Week 08 recommendation

Week 07 achieved its intended outcome: an executed, reproducible CNN baseline with fixed data preparation, validation-based checkpoint selection, saved metrics, a test confusion matrix, and qualitative failure evidence. The overall score improved over the strongest previous MLP under matching input geometry. Yet accuracy alone exaggerates practical reliability, because class-specific performance remains uneven and held-out scenes reduce performance substantially.

Suggested Week 08 plan - CNN Classification II

- Preserve the current CompactCNN as the baseline; compare one stronger architecture or pretrained CNN under clearly recorded training conditions.
- Measure robustness by scene, bounding-box scale, and annotation visibility bins where the available BOP metadata supports those analyses; treat viewpoint and clutter analysis as qualitative unless measurable labels are defined.
- Prioritize master_chef_can -> tomato_soup_can and bleach_cleanser -> tomato_soup_can. Inspect diverse hard examples and distinguish small objects, partial visibility, pose, and appearance before proposing targeted augmentation.
- Choose Week 08 model changes using training and new development/validation experiments only. The existing held-out test results have now been viewed and must not be repeatedly used for model selection; use genuinely unseen scenes for a new final test when possible.

8. Evidence, reproducibility, and limitations

Source of reported numerical and visual evidence: uploaded executed notebook week07_CNN_classification_I(2).ipynb (cells A-I). The notebook logs seed 42, class IDs, scene IDs, configuration, epoch history, class metrics, and test error transitions. It writes split CSVs, predictions, training history, checkpoint weights, confusion matrices, class metrics, plots, and a final-result JSON when executed in its runtime; those separate output files were not part of the submitted notebook attachment.

Main Quest evidence: the notebook demonstrates CNN concepts (convolutions, spatial pooling, nonlinear activation, dropout, cross-entropy and validation-based training). A separate record of completed course lesson titles was not included, so formal lesson completion cannot be verified from this submission.